

Supplementary Information

A Comparative Study of Linear and Neural Network Models for Liver–Brain MicroRNA Co-Expression in Spaceflight-Exposed Mice

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Supplementary Table S1

Per-fold LOOCV mean squared error for the unregularized linear regression and the nested ridge regression, with the ridge regularization strength (α) selected by inner leave-one-out cross-validation within each outer training fold, after that fold's brain-target set (main-text Table 2) had already been selected from its 8 training samples. Underlies the aggregate values reported in main-text Table 1 and Section 3.1.

Fold	Sample	Linear reg. MSE	Ridge reg. MSE	Ridge α
1	FL-LAR-1	3.4600	0.2495	100.0
2	FL-LAR-13	0.4365	0.4161	0.001
3	FL-LAR-5	0.4902	0.6862	1.0
4	HC-LAR-1	0.6711	7.4239	100.0
5	HC-LAR-13	1.4678	1.4774	0.001
6	HC-LAR-5	0.5181	0.3468	1.0
7	VC-LAR-1	0.8763	1.8382	100.0
8	VC-LAR-13	0.4540	2.4525	1.0
9	VC-LAR-5	2.0658	2.0374	0.01

Source: reproducibility/tables/baseline_fold_metrics.csv.

Supplementary Table S2

MLP validation mean squared error (VMSE) for each of the five fixed random seeds used in the canonical reproducibility pipeline. Mean = 1.6489; SD = 0.1396; range = 1.4527–1.8567. Underlies the seed-mean value reported in main-text Table 1, Figure 2, and Figure 3.

Seed	101	202	303	404	505
MLP VMSE	1.8567	1.5856	1.6027	1.4527	1.7467

Source: reproducibility/tables/metrics_by_seed.csv.

Supplementary Table S3

Per-fold linear-regression MSE paired against the five-seed-mean MLP MSE for that same held-out sample, underlying the fold-paired Wilcoxon signed-rank test reported in main-text Section 3.1. Signed

rank is assigned by absolute difference (rank 1 = smallest |difference|; no ties). A two-sided Wilcoxon signed-rank test on the nine signed differences gives $W = 19.0$, $p = 0.734$ (scipy.stats.wilcoxon, exact method, no ties); a sign test on the win/loss split (5 linear wins, 4 MLP wins) gives $p = 1.00$ (binomial, two-sided). Excluding fold 4 (HC-LAR-1), the mean fold-level MSE across the remaining eight folds is 1.2211 for linear regression and 1.0856 for the MLP. Derived from reproducibility/tables/baseline_fold_metrics.csv (linear) and reproducibility/tables/fold_metrics.csv (MLP, averaged across the five fixed seeds per fold); no additional model retraining was performed.

Fold	Sample	Linear MSE	MLP mean MSE	Diff (MLP-Linear)	Signed rank	Winner
1	FL-LAR-1	3.4600	0.7724	-2.6876	-8	MLP
2	FL-LAR-13	0.4365	0.2267	-0.2098	-4	MLP
3	FL-LAR-5	0.4902	0.3860	-0.1042	-2	MLP
4	HC-LAR-1	0.6711	6.1552	+5.4841	+9	Linear
5	HC-LAR-13	1.4678	1.6377	+0.1699	+3	Linear
6	HC-LAR-5	0.5181	0.2462	-0.2719	-5	MLP
7	VC-LAR-1	0.8763	0.9561	+0.0798	+1	Linear
8	VC-LAR-13	0.4540	1.5080	+1.0540	+7	Linear
9	VC-LAR-5	2.0658	2.9514	+0.8856	+6	Linear

"Diff" is MLP mean MSE minus linear MSE; a negative value means the MLP achieved the lower (better) MSE for that fold. "Winner" indicates which model achieved the lower fold-level MSE for that sample: linear wins 5 of 9 folds, the MLP wins 4 of 9. Source: reproducibility/tables/baseline_fold_metrics.csv, reproducibility/tables/fold_metrics.csv.